

THE RELATION BETWEEN THE NUMBER OF SPECIES AND
THE NUMBER OF INDIVIDUALS IN A RANDOM SAMPLE
OF AN ANIMAL POPULATION

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(With 8 Figures in the Text)

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PART I. RESULTS

By

It is well known that the distribution of species in a biological measurements usually falls into three types:

(a) the binomial distribution, in which the species are represented by the binomial $(q + p)^n$;

(b) the normal distribution, in which the species are distributed symmetrically about an average value, and which is the case when p and q are equal;

(c) the Poisson series, in which the species are expressed by the series

$$e^{-m} \left(1 + m + \frac{m^2}{2!} + \dots \right)$$

where m is the mean and e is the base of the logarithmic system, $e = 2.7183$.



Fisher, Corbet
and Williams
(1943)

Fisher's log-
series model

Expected number
of species appearing r times = $\alpha \times \frac{x^r}{r}$

Constant derived from the data

N. of species

Sample size

$$Y^{(n)} = \alpha \log \left(1 + \frac{n}{\alpha} \right)$$